	Design	Composition		Networks			Statistical information			Numerical conditioning			
Scenario	Genes G	Shannon $H^\star$	$\|\boldsymbol{p}\|_2^2$	Edge density	Mean degree	Hoyer sparsity	Mean overlap $\bar{\omega}$	Hellinger	max f_{cov}	$\kappa(\boldsymbol{\mu})$	$\lambda_{\min}(\boldsymbol{\mu}^\top \boldsymbol{\mu})$	$\kappa\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$	$\lambda_{\min}\{\boldsymbol{\Sigma}(\boldsymbol{p})\}$
balanced / low overlap													
CT1 Scale-free / CT2 Scale-free	20	1	0.333	0.1	1.9	0.329	0.0203	0.971	0.387	4.38	10	5.49	0.463
CT1 SBM / CT2 Scale-free	20	1	0.333	0.0912	1.73	0.32	0.0208	0.971	0.463	4.38	10	3.77	0.491
CT1 Scale-free / CT2 SBM	20	1	0.333	0.0982	1.87	0.304	0.0206	0.971	0.442	4.38	10	3.18	0.544
CT1 SBM / CT2 SBM	20	1	0.333	0.0895	1.7	0.376	0.0195	0.971	0.412	4.38	10	3.56	0.495
balanced / moderate overlap													
CT1 Scale-free / CT2 Scale-free	20	1	0.333	0.1	1.9	0.329	0.0765	0.907	0.718	4.38	10	5.49	1.86
CT1 SBM / CT2 Scale-free	20	1	0.333	0.0912	1.73	0.32	0.0802	0.897	0.833	4.38	10	3.77	2.84
CT1 Scale-free / CT2 SBM	20	1	0.333	0.0982	1.87	0.304	0.0817	0.895	0.812	4.38	10	3.18	2.96
CT1 SBM / CT2 SBM	20	1	0.333	0.0895	1.7	0.376	0.0802	0.9	0.756	4.38	10	3.56	2.18
balanced / high overlap													
CT1 Scale-free / CT2 Scale-free	20	1	0.333	0.1	1.9	0.329	0.2	0.769	0.926	4.38	10	5.49	9.19
CT1 SBM / CT2 Scale-free	20	1	0.333	0.0912	1.73	0.32	0.194	0.764	0.968	4.38	10	3.77	17.4
CT1 Scale-free / CT2 SBM	20	1	0.333	0.0982	1.87	0.304	0.198	0.769	0.957	4.38	10	3.18	15.4
CT1 SBM / CT2 SBM	20	1	0.333	0.0895	1.7	0.376	0.199	0.764	0.944	4.38	10	3.56	11.8
moderately unbalanced / low overlap													
CT1 Scale-free / CT2 Scale-free	20	0.5	0.719	0.1	1.9	0.355	0.0354	0.971	0.71	4.38	10	9.48	0.769
CT1 SBM / CT2 Scale-free	20	0.5	0.719	0.0912	1.73	0.606	0.041	0.971	0.794	4.38	10	7.39	0.887
CT1 Scale-free / CT2 SBM	20	0.5	0.719	0.0982	1.87	0.354	0.0369	0.971	0.764	4.38	10	9.13	0.811
CT1 SBM / CT2 SBM	20	0.5	0.719	0.0895	1.7	0.608	0.0406	0.971	0.78	4.38	10	7.39	0.911
moderately unbalanced / moderate overlap													
CT1 Scale-free / CT2 Scale-free	20	0.5	0.719	0.1	1.9	0.355	0.132	0.907	0.908	4.38	10	9.48	3.1
CT1 SBM / CT2 Scale-free	20	0.5	0.719	0.0912	1.73	0.606	0.157	0.897	0.957	4.38	10	7.39	5.13
CT1 Scale-free / CT2 SBM	20	0.5	0.719	0.0982	1.87	0.354	0.143	0.895	0.946	4.38	10	9.13	4.42
CT1 SBM / CT2 SBM	20	0.5	0.719	0.0895	1.7	0.608	0.151	0.9	0.94	4.38	10	7.39	4.02
moderately unbalanced / high overlap													
CT1 Scale-free / CT2 Scale-free	20	0.5	0.719	0.1	1.9	0.355	0.316	0.769	0.98	4.38	10	9.48	15.3
CT1 SBM / CT2 Scale-free	20	0.5	0.719	0.0912	1.73	0.606	0.341	0.764	0.993	4.38	10	7.39	31.5
CT1 Scale-free / CT2 SBM	20	0.5	0.719	0.0982	1.87	0.354	0.301	0.769	0.989	4.38	10	9.13	23
CT1 SBM / CT2 SBM	20	0.5	0.719	0.0895	1.7	0.608	0.336	0.764	0.988	4.38	10	7.39	21.8
highly unbalanced / low overlap													
CT1 Scale-free / CT2 Scale-free	20	0.1	0.961	0.1	1.9	0.355	0.0756	0.971	0.755	4.38	10	9.78	1.01
CT1 SBM / CT2 Scale-free	20	0.1	0.961	0.0912	1.73	0.613	0.1	0.971	0.824	4.38	10	7.58	1.17
CT1 Scale-free / CT2 SBM	20	0.1	0.961	0.0982	1.87	0.355	0.0836	0.971	0.81	4.38	10	9.77	1.03
CT1 SBM / CT2 SBM	20	0.1	0.961	0.0895	1.7	0.613	0.095	0.971	0.812	4.38	10	7.58	1.2
highly unbalanced / moderate overlap													
CT1 Scale-free / CT2 Scale-free	20	0.1	0.961	0.1	1.9	0.355	0.221	0.907	0.926	4.38	10	9.78	4.06
CT1 SBM / CT2 Scale-free	20	0.1	0.961	0.0912	1.73	0.613	0.288	0.897	0.964	4.38	10	7.58	6.78
CT1 Scale-free / CT2 SBM	20	0.1	0.961	0.0982	1.87	0.355	0.238	0.895	0.959	4.38	10	9.77	5.6
CT1 SBM / CT2 SBM	20	0.1	0.961	0.0895	1.7	0.613	0.271	0.9	0.95	4.38	10	7.58	5.3
highly unbalanced / high overlap													
CT1 Scale-free / CT2 Scale-free	20	0.1	0.961	0.1	1.9	0.355	0.477	0.769	0.984	4.38	10	9.78	20
CT1 SBM / CT2 Scale-free	20	0.1	0.961	0.0912	1.73	0.613	0.544	0.764	0.994	4.38	10	7.58	41.6
CT1 Scale-free / CT2 SBM	20	0.1	0.961	0.0982	1.87	0.355	0.448	0.769	0.992	4.38	10	9.77	29.1
CT1 SBM / CT2 SBM	20	0.1	0.961	0.0895	1.7	0.613	0.52	0.764	0.99	4.38	10	7.58	28.7